COM6006 Programming Concepts - 09 Python Recap

09 Data Libraries - Numpy, Pandas, Seaborn

Plan for this lecture

1. Matrices and Computer Vision
2. Numpy
3. Reading Data Files (CSVs) - Pandas
4. Data Visualisation (Seaborn / Matplotlib)
5. AI examples - iRaces AI?

0.0 The importance of matrices for Computer Vision!

- Images are 2D arrays of numbers!
- Grayscale = one 'channel' of 0 to 255 values.
- 0 = White
- 255 = Black
- RGB = three 'channels' - Red, Green and Blue, e.g. [R, 255, 0]

0.1 RGB Colour = 3 matrices

- RGB = Three 'channels' - Red, Green and Blue, e.g. [R, 255, 0]

0.2 Colour Wheels

- https://www.schemecolor.com/analogous_pink-blue.php
- https://RGBcolorpicker.com/0-ff-ff-0#66ff
- If you're dabbled in choosing colours for CSS or perhaps in Photoshop, you may have seen these hexadecimal codes for each colour.
- This is hexadecimal equivalent (base 16) of our decimal values (base 10) for RGB.

In [1]:
```
# Hex
# R, G, B
# 00, FF, 77
[R, 255, 0]
# Three hairy bits to a byte?
```

Out[1]:
```
[R, 255, 0]
```

In [2]:
```
"00FF"
```

Out[2]:
```
'00FF'
```

In [3]:
```
bin(255)
```

Out[3]:
```
'0b11111111'
```

In [4]:
```
bin(0)
```

Out[4]:
```
'0b0'
```

In [5]:
```
bin(255)[2:]
```

Out[5]:
```
'11111111'
```

In [6]:
```
bin(0)[2:] + bin(255)[2:] + bin(0)
```

Out[6]:
```
'0b00ffff00'
```

In [7]:
```
bin(255)[2:]*3
```

Out[7]:
```
'111111111111111111111111'
```

In [8]:
```
bin(255)[2:] + bin(255)[2:] + bin(255)[2:]
```

Out[8]:
```
'111111111111111111111111'
```

- 'FFFFFF'
  - So therefore, each pixel of a full-map images is at least 3 bytes...
  - Hairy bits, yeah! 8 x 3 = 24 pixel image!

0.3 Image file size

In [9]:
```
10 * 10
```

Out[9]:
```
100
```

- To be colour, we'd have to multiply this by 3 channels (RGB).

In [10]:
```
10 * 10 * 3
```

Out[10]:
```
300
```

- How about a HD 1080 x 1920 pixel image?

In [11]:
```
1080 * 1920
```

Out[11]:
```
2073600
```

In [12]:
```
1080 * 1920 * 3
```

Out[12]:
```
6220800
```

- 2 million pixels at 3 DP megapixels (black and white)
  - Or multiplied by 3 channels (RGB) = 6.2 million pixels (or 6.22 megapixels) mb
  - This is our uncompressed colour library (.BMP).
  - The Joint Photographic Group (.JPG) came up with a compression algorithm to reduce this file size down.

JPEG Compression

- JPEG uses lossy compression, which discards some visual information to shrink the file
- But this lossy compression does not actually lead to noticeable quality loss to the human eye.

Typical compression ratios:

- High quality: 1:60 - 1:80
- Medium quality: 1:80 - 1:90 - 1:95
- Low quality: > 1:95-1:99

So a 1080p image that is 6.2MB uncompressed might shrink to ~310 KB-1 MB as a typical JPEG! - that's about 5-10x smaller.

0.4 Game preparation (Exercise 10)

- Can you write code to recreate this pyramid with characters (e.g. '#')?

- How many dimensions do you need?
- Is there a convenient way to draw this blocks of the pyramid?

In [13]:
```
[ [ ], [ ], [ ], [ ], [ ], [ ], [ ], [ ] ]
```

Out[13]:
```
[[], [], [], [], [], [], [], []]
```

In [14]:
```
[ [ ], [ ], [ ], [ ], [ ], [ ], [ ], [ ] ]
```

Out[14]:
```
[[], [], [], [], [], [], [], []]
```

In [15]:
```
[ [ ], [ ], [ ], [ ], [ ], [ ], [ ], [ ] ]
```

Out[15]:
```
[[], [], [], [], [], [], [], []]
```

In [16]:
```
[ [ ], [ ], [ ], [ ], [ ], [ ], [ ], [ ] ]
```

Out[16]:
```
[[], [], [], [], [], [], [], []]
```

In [17]:
```
[ [ ], [ ], [ ], [ ], [ ], [ ], [ ], [ ] ]
```

Out[17]:
```
[[], [], [], [], [], [], [], []]
```

In [18]:
```
[ [ ], [ ], [ ], [ ], [ ], [ ], [ ], [ ] ]
```

Out[18]:
```
[[], [], [], [], [], [], [], []]
```

In [19]:
```
board = [[]]
```

Out[19]:
```
[[]]
```

Iterating:

In [20]:
```
for i in range(10):
    print(i)
```

Out[20]:
```
0
1
2
3
4
5
6
7
8
9
```

In [21]:
```
board = []
for i in range(10):
    board.append([])
    for j in range(8):
        board[i].append([])
```

In [22]:
```
board
```

Out[22]:
```
[[[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []]]
```

Using list comprehension:

In [23]:
```
board = [[] for _ in range(10)] for _ in range(10)]
board
```

Out[23]:
```
[[[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []],
 [[], [], [], [], [], [], [], []]]
```

In [24]:
```
height = 5\nwidth = 8
board = []
for i in range(height):
    board.append([])
    for j in range(width):
        board[i].append('')
for row in board:
    print(row)
```

Out[24]:
```
['', '', '', '', '', '', '', '']
['', '', '', '', '', '', '', '']
['', '', '', '', '', '', '', '']
['', '', '', '', '', '', '', '']
['', '', '', '', '', '', '', '']
```

In [25]:
```
for row in board:
    print(' '.join(row))
```

Out[25]:
```
# # # # # # # #
# # # # # # # #
# # # # # # # #
# # # # # # # #
# # # # # # # #
```

1. Introduction to Numpy

- Numbers! Python (NumPy) is a package full of methods that can perform useful operations on data.
- NumPy provides a convenient API (Application Programmable Interface) that provides a way to 'interface' with / operate on data.
- It deliberately uses arrays to make coding as over efficient way to store/transform data than the 'lousy' input values of Python that we've seen so far.
- More documentation available at https://numpy.org

1.1 Getting started with Numpy

- Type further search-based help if you are in VSC.
- Off if you're in Anaconda or Google Colab, you should have access to it already...

```
!pip install numpy
pip install numpy
jupyterlab-nb pip install[-10] numpy [-lab]
```

In [26]:
```
import numpy as np
```

In [27]:
```
numpy.__file__
```

Out[27]:
```
'/usr/local/lib/python3.9/site-packages/numpy/__init__.cp39-win_amd64.pyd'
```

- Let's compare the Numpy array with a Python List

1.2 Numpy Arrays vs Python List

- NumPy arrays are different to Python Lists.
- NumPy arrays restrictions the 'types' nature of more 'rectilinear' languages (C, C++, Java), where everything is explicitly typed.
- NumPy arrays operate the arrays from C and Java where they declare to store some type (ints, strings), utilise Python and JIL which can store data of different types.
- NumPy arrays therefore store a 'type' - floating point numbers to integers, or in some cases - an error is produced (strings to integers).

In [28]:
```
[1, 2, 3, 4, 5, 6, 7, 8]
```

Out[28]:
```
[1, 2, 3, 4, 5, 6, 7, 8]
```

In [29]:
```
numpy.array([1, 2, 3, 4, 5, 6, 7, 8])
```

Out[29]:
```
array([1, 2, 3, 4, 5, 6, 7, 8])
```

Mixing data types!

In [30]:
```
[1, 2, 3, 4, 5, 6, "Four", True]
```

Out[30]:
```
[1, 2, 3, 4, 5, 6, 'Four', True]
```

In [31]:
```
numpy.array([1, 2, 3, 4, 5, 6, "Four", True])
```

Out[31]:
```
array([1, 2, 3, 4, 5, 6, 'Four', 1])
```

- Numpy array DIE! - 'Four', 'True', 'Work'!), 'Work'!)

In [32]:
```
numpy.array([1, 2, 3, 4, 5, 6, "Four", True, "Work"])
```

Out[32]:
```
array([1, 2, 3, 4, 5, 6, 'Four', 'True', 'Work'])
```

In [33]:
```
numpy.array([1, 2, 3, 4, 5, 6, "Four", True, "Work", 7])
```

Out[33]:
```
array([1, 2, 3, 4, 5, 6, 'Four', '1', 'Work', '7'])
```

1.4 2-Dimensions!

In [34]:
```
numpy.array([1,2,3,4,5,6,7,8])
```

Out[34]:
```
array([1, 2, 3, 4, 5, 6, 7, 8])
```

In [35]:
```
numpy.array([1,2,3,4,5,6,7,8], ndmin=2)
```

Out[35]:
```
array([[1, 2, 3, 4, 5, 6, 7, 8]])
```

In [36]:
```
numpy.array([1, 2, 3, 4, 5, 6, 7, 8], ndmin=3)
```

Out[36]:
```
array([[[1, 2, 3, 4, 5, 6, 7, 8]]])
```

1.5 Array functions to assign values

In [37]:
```
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
```

Out[37]:
```
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
```

In [38]:
```
for i in range(0,10):
    print(i)
```

Out[38]:
```
0
1
2
3
4
5
6
7
8
9
```

In [39]:
```
numpy.zeros(10)
```

Out[39]:
```
array([0., 0., 0., 0., 0., 0., 0., 0., 0., 0.])
```

In [40]:
```
numpy.zeros(5, dtype=int)
```

Out[40]:
```
array([0, 0, 0, 0, 0])
```

In [41]:
```
numpy.ones(10)
```

Out[41]:
```
array([1., 1., 1., 1., 1., 1., 1., 1., 1., 1.])
```

In [42]:
```
numpy.ones(5, dtype=int)
```

Out[42]:
```
array([1, 1, 1, 1, 1])
```

1.6 Array attributes

In [43]:
```
numpy.random.seed(seed=1, size=5,5)
print("array1", numpy.ndim(array1))
print("array1", numpy.shape(array1))
print("array1", numpy.size(array1))
print("array1", numpy.dtype(array1))
size= 5
dtype= int
dimensions= 2

numpy.random.seed(seed=1, size=5,5,5)
print("array2", numpy.ndim(array2))
print("array2", numpy.shape(array2))
print("array2", numpy.size(array2))
print("array2", numpy.dtype(array2))
size= 5
dtype= int
dimensions= 3
```

Array slicing [: :]

```
array_name[ start : stop : interval]
```

In [44]:
```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

Out[44]:
```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

In [45]:
```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

Out[45]:
```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

In [46]:
```
numpy.array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

Out[46]:
```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

In [47]:
```
numpy.array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])[1:]
```

Out[47]:
```
array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

In [48]:
```
numpy.array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])[1:8]
```

Out[48]:
```
array([1, 2, 3, 4, 5, 6, 7])
```

In [49]:
```
numpy.array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])[1:8:2]
```

Out[49]:
```
array([1, 3, 5, 7])
```

In [50]:
```
numpy.array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

Out[50]:
```
array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

In [51]:
```
numpy.array([1, 2, 3, 4, 5, 6, 7, 8, 9])[:3]
```

Out[51]:
```
array([1, 2, 3])
```

In [52]:
```
numpy.array([1, 2, 3, 4, 5, 6, 7, 8, 9])[3:]
```

Out[52]:
```
array([4, 5, 6, 7, 8, 9])
```

Row and Col access

In [53]:
```
numpy.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
```

Out[53]:
```
array([[1, 2, 3],
       [4, 5, 6],
       [7, 8, 9]])
```

All of the first column (column 0) - displayed in a one dimensional form

In [54]:
```
numpy.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])[:,0]
```

Out[54]:
```
array([1, 4, 7])
```

All of the first row (row 0)

In [55]:
```
numpy.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])[0,:]
```

Out[55]:
```
array([1, 2, 3])
```

All of the second row (row 1)

In [56]:
```
numpy.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])[1,:]
```

Out[56]:
```
array([4, 5, 6])
```

Importing / Reading from files (CSVs)

- https://gamma.cs.unc.edu/FREE/iris-data/iris.data#iris.name

Iris Versicolor **Iris Setosa** **Iris Virginica**

In [57]:
```
import numpy as np
```

In [58]:
```
data = "https://archive.ics.uci.edu/ml/machine-learning-databases/iris/iris.data"
# Load the data, specify the delimiter
names = ["sepal_length", "sepal_width", "petal_length", "petal_width", "species"]
iris = numpy.genfromtxt(data, dtype=None, delimiter=",", names=names)
print(iris)
```

Out[58]:
```
[(5.1, 3.5, 1.4, 0.2, b'Iris-setosa')
 (4.9, 3.0, 1.4, 0.2, b'Iris-setosa')
 (4.7, 3.2, 1.3, 0.2, b'Iris-setosa')
 (4.6, 3.1, 1.5, 0.2, b'Iris-setosa')
 (5.0, 3.6, 1.4, 0.2, b'Iris-setosa')
 (5.4, 3.9, 1.7, 0.4, b'Iris-setosa')
 (4.6, 3.4, 1.4, 0.3, b'Iris-setosa')
 (5.0, 3.4, 1.5, 0.2, b'Iris-setosa')
 (4.4, 2.9, 1.4, 0.2, b'Iris-setosa')
 (4.9, 3.1, 1.5, 0.1, b'Iris-setosa')
 (5.4, 3.7, 1.5, 0.2, b'Iris-setosa')
 (4.8, 3.4, 1.6, 0.2, b'Iris-setosa')
 (4.8, 3.0, 1.4, 0.1, b'Iris-setosa')
 (4.3, 3.0, 1.1, 0.1, b'Iris-setosa')
 (5.8, 4.0, 1.2, 0.2, b'Iris-setosa')
 (5.7, 4.4, 1.5, 0.4, b'Iris-setosa')
 (5.4, 3.9, 1.3, 0.4, b'Iris-setosa')
 (5.1, 3.5, 1.4, 0.3, b'Iris-setosa')
 (5.7, 3.8, 1.7, 0.3, b'Iris-setosa')
 (5.1, 3.8, 1.5, 0.3, b'Iris-setosa')
 (5.4, 3.4, 1.7, 0.2, b'Iris-setosa')
 (5.1, 3.7, 1.5, 0.4, b'Iris-setosa')
 (4.6, 3.6, 1.0, 0.2, b'Iris-setosa')
 (5.1, 3.3, 1.7, 0.5, b'Iris-setosa')
 (4.8, 3.4, 1.9, 0.2, b'Iris-setosa')
 (5.0, 3.0, 1.6, 0.2, b'Iris-setosa')
 (5.0, 3.4, 1.6, 0.4, b'Iris-setosa')
 (5.2, 3.5, 1.5, 0.2, b'Iris-setosa')
 (5.2, 3.4, 1.4, 0.2, b'Iris-setosa')
 (4.7, 3.2, 1.6, 0.2, b'Iris-setosa')
 (4.8, 3.1, 1.6, 0.2, b'Iris-setosa')
 (5.4, 3.4, 1.5, 0.4, b'Iris-setosa')
 (5.2, 4.1, 1.5, 0.1, b'Iris-setosa')
 (5.5, 4.2, 1.4, 0.2, b'Iris-setosa')
 (4.9, 3.1, 1.5, 0.1, b'Iris-setosa')
 (5.0, 3.2, 1.2, 0.2, b'Iris-setosa')
 (5.5, 3.5, 1.3, 0.2, b'Iris-setosa')
 (4.9, 3.6, 1.4, 0.1, b'Iris-setosa')
 (4.4, 3.0, 1.3, 0.2, b'Iris-setosa')
 (5.1, 3.4, 1.5, 0.2, b'Iris-setosa')
 (5.0, 3.5, 1.3, 0.3, b'Iris-setosa')
 (4.5, 2.3, 1.3, 0.3, b'Iris-setosa')
 (4.4, 3.2, 1.3, 0.2, b'Iris-setosa')
 (5.0, 3.5, 1.6, 0.6, b'Iris-setosa')
 (5.1, 3.8, 1.9, 0.4, b'Iris-setosa')
 (4.8, 3.0, 1.4, 0.3, b'Iris-setosa')
 (5.1, 3.8, 1.6, 0.2, b'Iris-setosa')
 (4.6, 3.2, 1.4, 0.2, b'Iris-setosa')
 (5.3, 3.7, 1.5, 0.2, b'Iris-setosa')
 (5.0, 3.3, 1.4, 0.2, b'Iris-setosa')
 (7.0, 3.2, 4.7, 1.4, b'Iris-versicolor')
 (6.4, 3.2, 4.5, 1.5, b'Iris-versicolor')
 (6.9, 3.1, 4.9, 1.5, b'Iris-versicolor')
 (5.5, 2.3, 4.0, 1.3, b'Iris-versicolor')
 (6.5, 2.8, 4.6, 1.5, b'Iris-versicolor')
 (5.7, 2.8, 4.5, 1.3, b'Iris-versicolor')
 (6.3, 3.3, 4.7, 1.6, b'Iris-versicolor')
 (4.9, 2.4, 3.3, 1.0, b'Iris-versicolor')
 (6.6, 2.9, 4.6, 1.3, b'Iris-versicolor')
 (5.2, 2.7, 3.9, 1.4, b'Iris-versicolor')
 (5.0, 2.0, 3.5, 1.0, b'Iris-versicolor')
 (5.9, 3.0, 4.2, 1.5, b'Iris-versicolor')
 (6.0, 2.2, 4.0, 1.0, b'Iris-versicolor')
 (6.1, 2.9, 4.7, 1.4, b'Iris-versicolor')
 (5.6, 2.9, 3.6, 1.3, b'Iris-versicolor')
 (6.7, 3.1, 4.4, 1.4, b'Iris-versicolor')
 (5.6, 3.0, 4.5, 1.5, b'Iris-versicolor')
 (5.8, 2.7, 4.1, 1.0, b'Iris-versicolor')
 (6.2, 2.2, 4.5, 1.5, b'Iris-versicolor')
 (5.6, 2.5, 3.9, 1.1, b'Iris-versicolor')
 (5.9, 3.2, 4.8, 1.8, b'Iris-versicolor')
 (6.1, 2.8, 4.0, 1.3, b'Iris-versicolor')
 (6.3, 2.5, 4.9, 1.5, b'Iris-versicolor')
 (6.1, 2.8, 4.7, 1.2, b'Iris-versicolor')
 (6.4, 2.9, 4.3, 1.3, b'Iris-versicolor')
 (6.6, 3.0, 4.4, 1.4, b'Iris-versicolor')
 (6.8, 2.8, 4.8, 1.4, b'Iris-versicolor')
 (6.7, 3.0, 5.0, 1.7, b'Iris-versicolor')
 (6.0, 2.9, 4.5, 1.5, b'Iris-versicolor')
 (5.7, 2.6, 3.5, 1.0, b'Iris-versicolor')
 (5.5, 2.4, 3.8, 1.1, b'Iris-versicolor')
 (5.5, 2.4, 3.7, 1.0, b'Iris-versicolor')
 (5.8, 2.7, 3.9, 1.2, b'Iris-versicolor')
 (6.0, 2.7, 5.1, 1.6, b'Iris-versicolor')
 (5.4, 3.0, 4.5, 1.5, b'Iris-versicolor')
 (6.0, 3.4, 4.5, 1.6, b'Iris-versicolor')
 (6.7, 3.1, 4.7, 1.5, b'Iris-versicolor')
 (6.3, 2.3, 4.4, 1.3, b'Iris-versicolor')
 (5.6, 3.0, 4.1, 1.3, b'Iris-versicolor')
 (5.5, 2.5, 4.0, 1.3, b'Iris-versicolor')
 (5.5, 2.6, 4.4, 1.2, b'Iris-versicolor')
 (6.1, 3.0, 4.6, 1.4, b'Iris-versicolor')
 (5.8, 2.6, 4.0, 1.2, b'Iris-versicolor')
 (5.0, 2.3, 3.3, 1.0, b'Iris-versicolor')
 (5.6, 2.7, 4.2, 1.3, b'Iris-versicolor')
 (5.7, 3.0, 4.2, 1.2, b'Iris-versicolor')
 (5.7, 2.9, 4.2, 1.3, b'Iris-versicolor')
 (6.2, 2.9, 4.3, 1.3, b'Iris-versicolor')
 (5.1, 2.5, 3.0, 1.1, b'Iris-versicolor')
 (5.7, 2.8, 4.1, 1.3, b'Iris-versicolor')
 (6.3, 3.3, 6.0, 2.5, b'Iris-virginica')
 (5.8, 2.7, 5.1, 1.9, b'Iris-virginica')
 (7.1, 3.0, 5.9, 2.1, b'Iris-virginica')
 (6.3, 2.9, 5.6, 1.8, b'Iris-virginica')
 (6.5, 3.0, 5.8, 2.2, b'Iris-virginica')
 (7.6, 3.0, 6.6, 2.1, b'Iris-virginica')
 (4.9, 2.5, 4.5, 1.7, b'Iris-virginica')
 (7.3, 2.9, 6.3, 1.8, b'Iris-virginica')
 (6.7, 2.5, 5.8, 1.8, b'Iris-virginica')
 (7.2, 3.6, 6.1, 2.5, b'Iris-virginica')
 (6.5, 3.2, 5.1, 2.0, b'Iris-virginica')
 (6.4, 2.7, 5.3, 1.9, b'Iris-virginica')
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 (6.5, 3.0, 5.5, 1.8, b'Iris-virginica')
 (7.7, 3.8, 6.7, 2.2, b'Iris-virginica')
 (7.7, 2.6, 6.9, 2.3, b'Iris-virginica')
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 (6.9, 3.2, 5.7, 2.3, b'Iris-virginica')
 (5.6, 2.8, 4.9, 2.0, b'Iris-virginica')
 (7.7, 2.8, 6.7, 2.0, b'Iris-virginica')
 (6.3, 2.7, 4.9, 1.8, b'Iris-virginica')
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 (6.2, 2.8, 4.8, 1.8, b'Iris-virginica')
 (6.1, 3.0, 4.9, 1.8, b'Iris-virginica')
 (6.4, 2.8, 5.6, 2.1, b'Iris-virginica')
 (7.2, 3.0, 5.8, 1.6, b'Iris-virginica')
 (7.4, 2.8, 6.1, 1.9, b'Iris-virginica')
 (7.9, 3.8, 6.4, 2.0, b'Iris-virginica')
 (6.4, 2.8, 5.6, 2.2, b'Iris-virginica')
 (6.3, 2.8, 5.1, 1.5, b'Iris-virginica')
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 (6.7, 3.1, 5.6, 2.4, b'Iris-virginica')
 (6.9, 3.1, 5.1, 2.3, b'Iris-virginica')
 (5.8, 2.7, 5.1, 1.9, b'Iris-virginica')
 (6.8, 3.2, 5.9, 2.3, b'Iris-virginica')
 (6.7, 3.3, 5.7, 2.5, b'Iris-virginica')
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 (6.3, 2.5, 5.0, 1.9, b'Iris-virginica')
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 (6.2, 3.4, 5.4, 2.3, b'Iris-virginica')
 (5.9, 3.0, 5.1, 1.8, b'Iris-virginica')]
```

```
[0.0, 0.0], 'Iris-setosa'],
[0.0, 0.0], 'Iris-setosa'],
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[0.0, 0.0], 'Iris-setosa'],
[0.0, 0.0], 'Iris-versicolor'],
[0.0, 0.0], 'Iris-versicolor'],
[0.0, 0.0], 'Iris-versicolor
```

In [5]: ```
iris_df.info()
```

Out[5]:
```
<class'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column   Non-Null Count Dtype
--- ------   -------------- -----
 0   sepal_length 150 non-null   float64
 1   sepal_width 150 non-null   float64
 2   petal_length 150 non-null   float64
 3   petal_width 150 non-null   float64
 4   species   150 non-null   object
dtypes: float64(4), object(1)
memory usage: 5.9 KB
```

In [6]: ```
iris_df.shape
```

Out[6]:
```
(150, 5)
```

In [7]: ```
iris_df.columns
```

Out[7]:
```
Index(['sepal_length', 'sepal_width', 'petal_length',
       'petal_width', 'species'],
      dtype='object')
```

In [8]: ```
iris_df.dtypes
```

Out[8]:
```
sepal_length    float64
sepal_width     float64
petal_length    float64
petal_width     float64
species         object
dtype: object
```

In [9]: ```
iris_df.nunique()
```

Out[9]:
```
sepal_length   35
sepal_width   23
petal_length   43
petal_width   22
species   3
dtype: int64
```

In [10]: ```
iris_df.isnull()
```

Out[10]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
145	False	False	False	False	False
146	False	False	False	False	False
147	False	False	False	False	False
148	False	False	False	False	False
149	False	False	False	False	False

150 rows × 5 columns

In [11]: ```
iris_df.isnull().sum()
```

Out[11]:
```
sepal_length   0
sepal_width   0
petal_length   0
petal_width   0
species   0
dtype: int64
```

In [12]: ```
iris_df.describe()
```

Out[12]:

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

In [13]: ```
iris_df["sepal_length".mean()
```

Out[13]:
```
5.843333333333333
```

In [14]: ```
iris_df["sepal_width".mean()
```

Out[14]:
```
3.1
```

In [15]: ```
iris_df["petal_length".mean()
```

Out[15]:
```
3.7580000000000002
```

In [16]: ```
iris_df["petal_width".mean()
```

Out[16]:
```
1.1993333333333333
```

Let's use Seaborn for data visualisation

```
pip install seaborn
```

```
pip install'numpy<2.0.0'seaborn'<1.0a0'
```

In [17]: ```
%pip install -q -U seaborn
```

In [18]: ```
import seaborn as sns
```

In [19]: ```
sns.histplot(iris_df, x="petal_length", y="petal_width", hue="species")
```

In [20]: ```
sns.violinplot(data=iris_df, x="petal_length", y="species")
```

In [21]: ```
sns.pairplot(iris_df, hue="species")
```

In [22]: ```
requests.exceptions.ReadTimeout: HTTPSConnectionPool(host='www.itunes.apple.com', port=443): Read timed out. (read timeout=15)
```

4.0 APIs and Data Science - iTunes example

Example searches:

- https://itunes.apple.com/search?term=jack+johnson
- https://itunes.apple.com/search?term=jack+johnson&limit=25
- https://itunes.apple.com/search?term=sequenzia&limit=25

Not just music either. Iframe like tracks, which you can lookup via ISRC:

- https://itunes.apple.com/lookup?isrc=USRC10600269

For more information on searching, refer to Apple's developer's guide, and the iTunes documentation here.

Depending on which environment you're running this notebook, you may need to install `requests` before using the package.

```
pip install requests
```

```
pip install -q -U requests --user
```

In [23]: ```
import requests
```

In [24]: ```
def search_itunes(entity, media_type="music"):
    """Search the iTunes API using the specified entity and media type.

    Keyword Arguments:
        entity (str): The search term.
        media_type (str): The type of media to search for (e.g., music, movie, podcast).

    Returns:
        dict: The JSON response from the iTunes API.

    base_url = "https://itunes.apple.com/search"
    params = {
        "term": entity,
        "media": media_type,
        "limit": 10  # Limit the number of results to 10
    }

    response = requests.get(base_url, params=params)

    # Check if the request was successful
    if response.status_code == 200:
        print("Successfully fetched data.")
        return response.json()
    else:
        print(f"Failed to fetch data. Status code: {response.status_code}")
        raise Exception("Failed to fetch data from iTunes API")

# Example usage
entity = "jack johnson"
media_type = "music"
data = search_itunes(entity, media_type)
print(data)
```

Out[24]:
```
Successfully fetched data.
{'resultCount': 10, 'results': [{'wrapperType': 'track', 'kind': 'song', 'artistId': 159260351, 'collectionId': 1440650428, 'trackId': 1440650431, 'artistName': 'Jack Johnson', 'collectionName': 'In Between Dreams (Bonus Track Version)', 'trackName': 'Jack Johnson', 'collectionCensoredName': 'In Between Dreams (Bonus Track Version)', 'trackCensoredName': 'Jack Johnson', 'artistViewUrl': 'https://music.apple.com/us/artist/jack-johnson/159260351?uo=4', 'collectionViewUrl': 'https://music.apple.com/us/album/jack-johnson/1440650428?i=1440650431&uo=4', 'trackViewUrl': 'https://music.apple.com/us/album/jack-johnson/1440650428?i=1440650431&uo=4', 'previewUrl': 'https://audio-ssl.itunes.apple.com/itunes-assets/AudioPreview125/v4/ed/ff/ae/edffae7f-2a7f-3a7f-5a8f-5a6a5111ccca.m4a', 'artwork100x100': 'https://is1-ssl.mzstatic.com/image/thumb/Music115/v4/ed/ff/ae/edffae7f-2a7f-3a7f-5a8f-5a6a5111ccca/mzi.ldzqqqqq.jpg/100x100bb.jpg', 'collectionPrice': 10.99, 'trackPrice': 1.29, 'releaseDate': 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In [25]: ```
data
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In [ ]: [print("Number of results: {}".format(len(results)))*10]
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Number of results: 20
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  'bigImageURL
```

Python Shorthand	Hackerrank Solutions	2016-01-18 00:00:00	2016-01-18 00:00:00	4.5	5.0	63.0
Linear 1 Classification	Machine Learning	2016-01-18 00:00:00	2016-01-18 00:00:00	4.5	5.0	63.0
Fundamentals of Programming using Python	Brave Ebrey	Very Python and why another Python course	2016-01-19 00:00:00	4.0	5.0	52
Python Programming	Ryan Tower	Isn't you looking for a complete Python course	2016-01-19 00:00:00	Nah	Nah	Nah
Python Programming	Brave Ebrey	Programming Geeks Python	2016-01-19 00:00:00	4.0	5.0	52
Python Programming	Knowledge Flow	★★★★★ (rotten) (1/5 of 5)	2016-01-19 00:00:00	Nah	Nah	Nah
Python Programming	Ryan Tower	Isn't you looking for a complete Python course	2016-01-19 00:00:00	Nah	Nah	Nah
Non-Programer's Tutorial for Python 3	Wikibooks	Here you can easily download your Python	2016-01-20 00:00:00	4.0	5.0	31.0
Learn Python	Bryan Bibeault	Following topics in Python programming	2016-01-20 00:00:00	4.0	5.0	31.0

```
In [1]: df_shape[df_shape['shape'].isin(['(4, 5, 5)'])]
```

	shape	description	relevantness	price	averagerating	averaltrajectory
0	Python for Beginners: A Concise and Clear Intro	Gerald Carter	This book is designed to introduce programming and	2016-01-16 00:00:00	4.5	5.0
1	Python 3 Tutorial	Dan Bader	This is an iBooks version of Python	2016-01-17 00:00:00	4.5	5.0
2	Fundamentals of Programming using Python 3	Brave Ebrey	Very Python and why another Python	2016-01-18 00:00:00	4.0	5.0
3	Non-Programer's Tutorial for Python 3	Wikibooks	Here you can easily download your	2016-01-20 00:00:00	4.0	31.0

This **Jupyter Notebook** contains exercises for you to extend your introduction to the basics with Python libraries and packages. Attempt the following exercises, which slowly build in complexity. If you get stuck, check back to the Python lecture recording on Computer Vision or the Python lecture slides on the NYU Tandon page of Python Tutorial, which includes examples, exercises and quizzes to help your understanding.

```
pip install numpy
pip install matplotlib
pip install pandas
pip install seaborn
pip install sklearn
```

Exercise 1

Import numpy as np, and create a `ones(5,5)` array with values 0 - 9

Is there a function that you can use to assign the range of values?

Hint: If you're in Visual Studio Code - you'll probably need to install the packages via the `pip install` commands above. These libraries should be pre-installed in Anaconda and Google Colab

```
In [ ]: import numpy as np ones
```

```
In [ ]: # Create your array here
```

Exercise 2

Create a `ones(5,5)` array of zeros.

```
In [ ]: # Create your array here.
```

Exercise 3

Create a `ones(5,5)` array of random values.

```
In [ ]: # Create your array here.
```

Exercise 4

Create a `ones(5,5)` array with values 0 - 10

```
In [ ]: # Write your solution here.
```

Exercise 5

Now create a `ones(5,5)` array with only odd numbers between range 0-10 and 10-20.

```
In [ ]: # Write your solution here.
```

Exercise 6

Create an array of the following two dimensional array (`dim=2`) into a two dimensional array (5 x 5).

```
In [1]: ten = np.arange(1,26)
ten
```

```
Out[1]: array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16, 17,
       18, 19, 20, 21, 22, 23, 24, 25])
```

```
In [ ]: # Write your solution here
```

```
Out[1]: array([[ 1,  2,  3,  4,  5],
       [ 6,  7,  8,  9, 10],
       [11, 12, 13, 14, 15],
       [16, 17, 18, 19, 20],
       [21, 22, 23, 24, 25]])
```

Exercise 7

Of the nonlinear two dimensional array, how would you print just the middle column?

```
In [ ]: # Write your solution here
```

```
Out[1]: array([ 3,  8, 13, 18, 23])
```

Exercise 8

Create a two dimensional `ones(5,5)` array - it should be three rows by three columns (3 x 3). The values should be between the integer values 0 and 9, allocated randomly.

Put this out a few times to see if the values in your 3D are generated randomly each time.

```
In [ ]: # Write your solution here.
```

Exercise 9

Starting with the below two dimensional array - see if you can use some exception handling code to respond to `IndexErrors`.

```
In [1]: two_d = np.array([[1,2,3,4,5], [6,7,8,9,10]])
two_d
```

```
Out[1]: array([[ 1,  2,  3,  4,  5],
       [ 6,  7,  8,  9, 10]])
```

Write some exception handling code to respond appropriately to index values below the lower bound, and values above the upper bound. For example below:

```
In [1]: two_d[3,5] # above the upper bound of the array
```

```
IndexError                                Traceback (most recent call last)
<ipykernel_launcher_zip_1>:1 in <module>
----> 1 two_d[3,5] # above the upper bound of the array

IndexError: index 3 is out of bounds for axis 0 with size 2
```

```
In [1]: two_d[-5,-1] # below the lower bound of the array
```

```
IndexError                                Traceback (most recent call last)
<ipykernel_launcher_zip_1>:1 in <module>
----> 1 two_d[-5,-1] # below the lower bound of the array

IndexError: index -5 is out of bounds for axis 0 with size 2
```

Exercise 10

In preparation for your game, write code to print the 8 `'.'` symbols arranged in a pyramid the below:

Extension: Can you put this in a function which can create the pyramid of any size?

```
In [ ]: # Write your solution here
```

```
In [ ]: # Write your solution here
```

```
In [ ]: # Write your solution here
```

```
  "wrapperType": "track",
  "kind": "song",
  "artistId": 159260351,
  "collectionId": 1468058165,
  "trackId": 1468058699,
  "artistName": "Taylor Swift",
  "collectionName": 'Lover',
  "trackName": 'Soon You'll Get Better (feat. The Chicks)',
  "collectionCensoredName": 'Lover',
  "trackCensoredName": 'Soon You'll Get Better (feat. The Chicks)',
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  "trackViewUrl": "https://music.apple.com/us/album/soon-you%ll-get-better-feat-the-chicks/1468058165?i=1468058699&uo=4",
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  "collectionName": 'Lover',
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  "collectionCensoredName": 'Lover',
  "trackCensoredName": 'Miss Americana & The Heartbreak Prince',
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  "trackViewUrl": "https://music.apple.com/us/album/miss-americana-the-heartbreak-prince/1468058165?i=1468058180&uo=4",
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  "trackCensoredName": 'I Forgot That You Existed',
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  "trackViewUrl": "https://music.apple.com/us/album/i-forgot-that-you-existed/1468058165?i=1468058169&uo=4",
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  "trackViewUrl": "https://music.apple.com/us/album/paper-rings/1468058165?i=1468058181&uo=4",
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  "collectionPrice": 11.99,
  "trackPrice": 1.29,
  "releaseDate": "2019-08-23T07:00:00Z",
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  "trackExplicitness": 'notExplicit',
  "discCount": 1,
  "discNumber": 1,
  "trackCount": 18,
  "trackNumber": 10,
  "trackTimeMillis": 198530,
  "country": 'USA',
  "currency": 'USD',
  "primaryGenreName": 'Pop',
  "isStreamable": True}])
```


See below the attempt to format the data returned from the `search_itunes()` function to format the data into a DataFrame.

Question: What do you notice about the results column below?

In [3…
```
#from search_itunes import search_itunes #only if you have a .py script for search_itunes

df = pd.DataFrame(search_itunes("Taylor", "music"))
df
```

Out[3…

	resultCount	results
0	10	{'wrapperType': 'track', 'kind': 'song', 'arti…
1	10	{'wrapperType': 'track', 'kind': 'song', 'arti…
2	10	{'wrapperType': 'track', 'kind': 'song', 'arti…
3	10	{'wrapperType': 'track', 'kind': 'song', 'arti…
4	10	{'wrapperType': 'track', 'kind': 'song', 'arti…
5	10	{'wrapperType': 'track', 'kind': 'song', 'arti…
6	10	{'wrapperType': 'track', 'kind': 'song', 'arti…
7	10	{'wrapperType': 'track', 'kind': 'song', 'arti…
8	10	{'wrapperType': 'track', 'kind': 'song', 'arti…
9	10	{'wrapperType': 'track', 'kind': 'song', 'arti…

Your task is to transform the above DataFrame into the format shown below:

In [3…
```
# Write your solution here...
```

Out[3…

wrapperType	kind	artistId	collectionId	trackId	artistName	collectionName	trackName	collectionCensoredName	trackCensoredName	…	di
track	song	400555769	1506831755	1506832357	Natalie Taylor	Surrender - Single	Surrender	Surrender - Single	Surrender	…	
track	song	400555769	1182561824	1182561980	Natalie Taylor	In the Air Tonight - Single	In the Air Tonight	In the Air Tonight - Single	In the Air Tonight	…	
track	song	400555769	880114237	880114240	Natalie Taylor	Love Life - Single	Love Life	Love Life - Single	Love Life	…	
track	song	400555769	1519882762	1519882765	Natalie Taylor	Surrender (Kina Remix) - Single	Surrender (Kina Remix)	Surrender (Kina Remix) - Single	Surrender (Kina Remix)	…	
track	song	400555769	1178290096	1178290111	Natalie Taylor	Loved You Before - Single	Loved You Before	Loved You Before - Single	Loved You Before	…	
track	song	645420096	1572859144	1572859302	ILLENIUM & Natalie Taylor	Fallen Embers	Fragments	Fallen Embers	Fragments	…	
track	song	400555769	978021284	978021290	Natalie Taylor	Wildfire - EP	Wildfire	Wildfire - EP	Wildfire	…	
track	song	400555769	1457378386	1457378387	Natalie Taylor	Iris - Single	Iris	Iris - Single	Iris	…	
track	song	400555769	1154603331	1154603480	Natalie Taylor	Amen - Single	Amen	Amen - Single	Amen	…	
track	song	400555769	988419300	988419846	Natalie Taylor	I Want It All - Single	I Want It All	I Want It All - Single	I Want It All	…	

10 rows × 32 columns

Extension: Write code which returns the song with the longest `trackTime`

Hint: Start by identifying the column that has data on the `trackTime`

In […

Exercise 16

Find an CSV file online, that you can import via `read_csv()`. Be aware that the data needs to be in raw format (without the html/css/table structure). Alternatively, download the CSV file and read it locally (as some sites won't give you the raw data in a readable/scrapable format).

Examples of CSV data:

https://github.com/riteshc6/amazon_scraper/blob/master/downloads/Computers%20%26%20Accessories.csv

https://raw.githubusercontent.com/riteshc6/amazon_scraper/master/downloads/Computers%20%26%20Accessories.csv

In [1…
```
url = "https://raw.githubusercontent.com/riteshc6/amazon_scraper/master/downloads/Computers%20%26%20Accessories.csv"

df = pd.read_csv(url)

df
```

Inspect the data from your chosen CSV, and perform some exploratory data analysis - use Seaborn and/or Matplotlib to visualise trends.

Question: What trends do you see?

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nbviewer GitHub repository.

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